SemantiCodec: An Ultra Low Bitrate Semantic Audio Codec for General Sound

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***Abstract*—Large language models (LLMs) have significantly advanced audio processing through audio codecs that convert audio into discrete tokens, enabling the application of language modelling techniques to audio data. However, traditional codecs often operate at high bitrates or within narrow domains such as speech and lack the semantic clues required for efficient language modelling. Addressing these challenges, we introduce SemantiCodec, a novel codec designed to compress audio into fewer than a hundred tokens per second across diverse audio types, including speech, general sound, and music, without compromising quality. SemantiCodec features a dual-encoder architecture: a semantic encoder using a self-supervised pre-trained Audio Masked Autoencoder (AudioMAE), discretized using k-means clustering on extensive audio data, and an acoustic encoder to capture the remaining details. The semantic and acoustic encoder outputs are used to reconstruct audio via a diffusion-model-based decoder. SemantiCodec is presented in three variants with token rates of 25, 50, and 100 per second, supporting a range of ultra-low bit rates between 0.31 kbps and 1.40 kbps. Experimental results demonstrate that SemantiCodec significantly outperforms the state-of-the-art Descript codec on reconstruction quality. Our results also suggest that SemantiCodec contains significantly richer semantic information than all evaluated state-of-the-art audio codecs, even at significantly lower bitrates. Our code and demos are available at https://haoheliu.github.io/SemantiCodec/.**

***Index Terms*—audio codec, semantic, low bitrate**

I. Introduction

AUDIO codecs are used for encoding and decoding digital audio for efficient telecommunications and broadcasting [1]. Traditional audio encoders compress audio by discarding inaudible details to reduce storage and transmission demands [1]. The degree of compression is typically assessed by the bitrate, indicating the amount of data, in bits per second, used to represent the audio signal, with commonly used bitrate ranges such as 128 kbps to 320 kbps for MP3 [2] and 6 kbps to 510 kbps for the Opus codec [3].

With the introduction of deep learning, audio codecs have significantly evolved with better audio quality and bitrate efficiency [4]. These cutting-edge codecs utilize vector quantization (VQ) [5] to learn compact codebooks, whose indices are transmitted instead of raw audio data. The sequence of transmitted indices is also referred to as the token sequence. Unlike traditional audio codecs, neural audio codecs typically operate at lower bit rates while maintaining similar audio quality. For instance, Encodec [6] achieves compression at multiple bitrates between 1.5 kbps and 24 kbps, the Descript codec [7] operates at 8 kbps and 16 kbps, and the HiFi-Codec pushes the boundaries further by reducing the bitrate to 2 kbps with acceptable quality [8].

Beyond the fundamental role of storing and transmitting audio, audio codecs have emerged as critical components in the domain of audio language modelling [9]. Similar to the tokenizers used in text processing [10], the neural audio codecs simplify complex audio waveforms into discrete integer tokens with substantially shorter lengths and align the training process of audio language models closely with the training methodologies of text-based language models [11]. This simplification has enabled models like AudioLM [9] to perform next-token prediction on audio codec sequences, demonstrating success in generating semantically plausible audio continuations. Further advancements have explored the incorporation of text as conditional information, which has paved the way for innovative applications such as text-to-audio generation with AudioGen [12], text-to-music generation through MusicLM [13] and MusicGen [14], and enhanced text-to-speech systems exemplified by VALL-E [15]. The integration of audio codecs into language models has also significantly advanced audio understanding capabilities, as demonstrated by the LTU model [16]. Moreover, developing systems like AudioPaLM [17] demonstrates the potential of the joint understanding and generation of speech, merging text-based and audio-based language models into a unified framework.

Despite the advancement in audio language modelling, the token rate of audio codecs has become a growing concern. For example, the token rate for a 6 kbps Descript codec is 600 per second (using six codebooks of size 1024). The autoregressive (AR) nature of token generation means that the inference time of an audio language model scales with the rate of codec tokens, posing challenges not only in computational efficiency but also in model training, where longer sequences demand more computational resources. Additionally, longer sequences may lead to challenges on long-term dependencies. Studies on long-context language models [18] indicate that language models do not effectively leverage long-term context, often superficially utilising it. While low-bitrate audio codecs are available, like the 1.5 kbps version of Encodec [6], their reconstruction quality, with strong artefacts introduced, often falls short of production standards. This situation underscores a crucial trade-off between efficiency and quality in audio codec development, highlighting the need for codecs that can achieve high-quality audio reconstruction at low bitrates.

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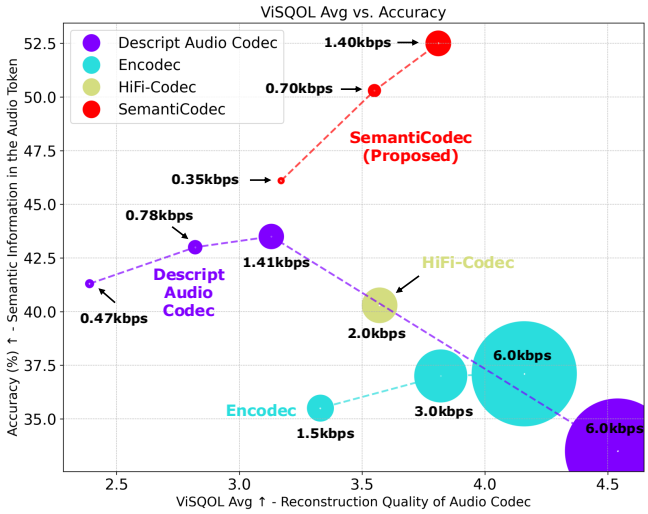

Fig. 1. Comparison between SemantiCodec and state-of-the-art codecs. Higher values on the horizontal and vertical axis indicate better reconstruction quality and semantic information, respectively. The size of circles indicates bitrates, where smaller ones denote lower bitrates.

The presence and richness of semantic information within token sequences play a crucial role in the learning of language models [19], [20]. Support for this assertion also comes from findings by Toraman et al. [10], who demonstrated that on six text processing tasks tokenizers operating at a higher level of granularity, such as byte pair encoding [21], could substantially outperform character-level tokenizers, which often require the model to expand capacity for understanding. Despite these insights on the importance of semantic information in the tokens, our initial investigations reveal that existing audio codecs, already employed in audio language modelling, fall short of capturing adequate semantic information, even at relatively high bitrate settings. For instance, when employing the latent encodings of the 6.0 kbps Descript codec [7] for classification tasks across seven benchmarks in the HEAR benchmark [22], the average accuracy is only 33%, as demonstrated in Section V-B. In contrast, without fine-tuning, a self-supervised pretrained AudioMAE encoder [23] results in a significantly higher accuracy of 61%. The classification accuracy is the indicator of semantic richness within codecs, which is crucial for audio language modelling. Moreover, our analysis indicates that using just the first vector quantization layer of the 6.0 kbps Descript codec, which is often considered to be the most important layer and selected for audio language modelling [15], the classification accuracy drops even further to an average of 25%. This deficiency in capturing and encoding rich semantic information in audio codecs can potentially hinder the performance of audio language models.

In this paper, we introduce SemantiCodec, a novel audio codec designed to tackle the issues of excessive token rate and insufficient semantic encoding in current audio codecs. SemantiCodec exhibits richer semantic information and similar reconstruction quality with lower bitrates than previous codecs, as illustrated in Figure 1. Our approach uses the strong generative capabilities of diffusion models alongside the rich audio representations learned by the self-supervised AudioMAE. SemantiCodec processes mel-spectrograms through two encoders sequentially with two distinct vector quantization (VQ) layers. The first VQ layer is constructed using centroids derived from k-means clustering [24] performed on a large dataset of AudioMAE features, ensuring the capture of semantic information. In contrast, the second layer employs a conventional learnable VQ mechanism, enhancing the audio reconstruction fidelity of SemantiCodec. Our empirical analysis reveals that the semantic content is predominantly encoded within the first VQ layer, contributing over 95% to the classification accuracy. However, integrating the second VQ layer is critical in significantly elevating the quality and intelligibility of the reconstructed audio. The concatenated quantized outputs from both VQ layers serve as the conditional input for a latent diffusion model (LDM), which follows the architecture of AudioLDM [25]. The latent diffusion model utilizes the provided conditions to reconstruct high-quality audio after quantization. Our experiment shows that SemantiCodec, at similar bitrates, outperforms the state-of-the-art audio codecs on reconstruction quality and contains significantly richer semantic information in the audio tokens, leading to improved classification accuracy in audio understanding. In summary, our contributions are listed as follows:

- We propose SemantiCodec, which leverages strong generative models and the rich features learnt by a self-supervised model for semantic-driven audio encoding and reconstruction.
- To our knowledge, SemantiCodec is the first work to use a diffusion based decoder in a neural audio codec at the time of writing.
- SemantiCodec achieves strong reconstruction performance across general sound types at exceptionally low token rates of 25, 50, and 100 per second, surpassing counterparts operating at significantly higher token rates.
- Evaluation on audio classification benchmarks demonstrates the significantly richer semantic information in the tokens of SemantiCodec, even with a single layer of vector quantization, indicating strong potential in future audio language modelling with the SemantiCodec.

II. Related Work

A. Neural Audio Codecs

Traditional audio codecs, as detailed by Valin et al. [26] and Dietz et al. [27], have demonstrated their ability to achieve low latency audio compression across a variety of audio types. However, these traditional approaches often fail to deliver high-quality audio reconstruction at low-bitrate settings, such as 3.0 kbps. Pioneering deep-learning-based audio codecs, Garbacea et al. [28] introduce the use of vector-quantized variational auto-encoders (VQ-VAEs) for learning neural codecs tailored to speech data. SoundStream [4] proposes a universal codec adaptable to various audio types, incorporating a residual vector quantization (RVQ) module to enhance the quantization process and a generative adversarial network (GAN) [29] to improve the reconstruction quality. Following a similar trajectory, Encodec [6] advances the capabilities of SoundStream

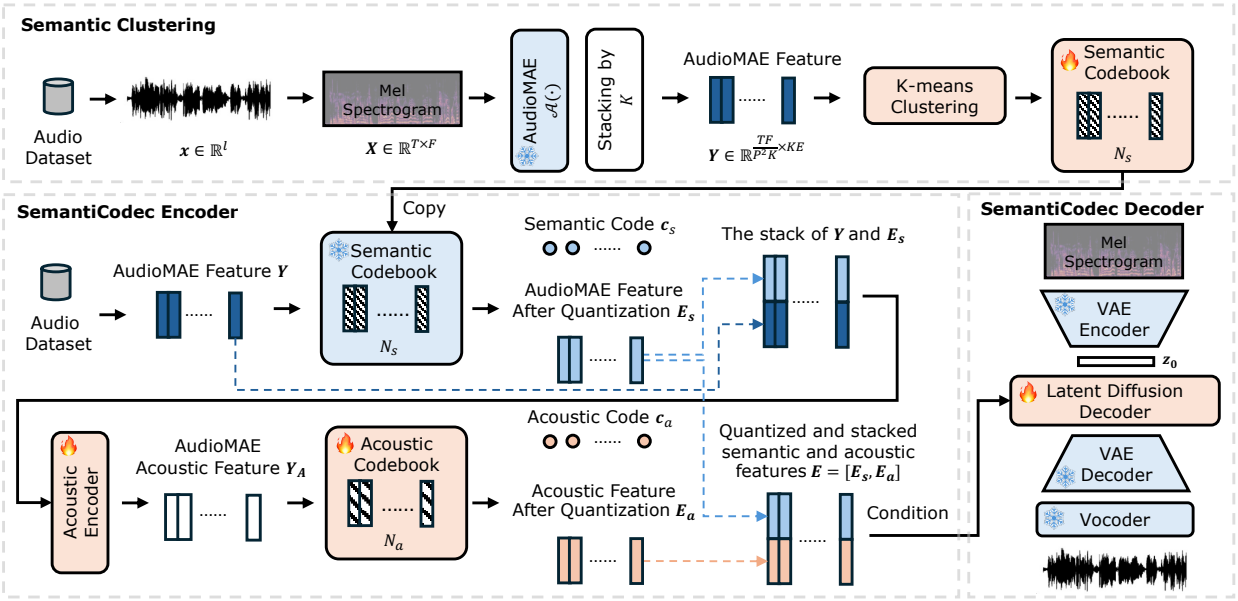

Fig. 2. SemantiCodec architecture. For an input audio clip, quantized semantic representation $\boldsymbol{E}_s$ is obtained via a pre-computed codebook using k-means clustering on the AudioMAE embeddings. Then $\boldsymbol{Y}$ and $\boldsymbol{E}_s$ are concatenated and fed to a residual encoder to complement acoustic details, which is discretized to $\boldsymbol{E}_a$ by a vector quantization module. SemantiCodec encoder output $\boldsymbol{E}$ is obtained by concatenating $\boldsymbol{E}_s$ and $\boldsymbol{E}_a$. A latent diffusion model is trained to generate the original audio clip conditioned on $\boldsymbol{E}$. The snowflake and fire symbols denote frozen and learnable parameters, respectively.

by integrating multi-scale discriminators and a loss-balancing strategy for reconstruction alongside an additional language model to facilitate further compression. HiFi-Codec [8] introduced group-residual vector quantization (GRVQ), a novel approach aimed at reducing the number of codebooks required while preserving the reconstruction quality. Descript codec [7] offers enhancements to Encodec, achieving significantly better reconstruction performance.

To compare our proposed SemantiCodec with previous systems, our experimental analysis primarily focuses on evaluating the richness of semantic information and the quality of audio reconstruction.

B. Semantic Audio Representation Learning

Self-supervised learning (SSL) has exhibited strong performance in audio representation learning. SSL models can be categorized into two types based on their pre-training tasks: discriminative SSL and reconstructive SSL. Discriminative SSL models, exemplified by HuBERT [30], COLA [31], and BEATs [32], employ strategies such as contrastive learning to differentiate between positive and negative pairs, or masked language modelling (MLM) techniques to predict the quantized labels of masked segments, leveraging contextual information. Conversely, reconstructive SSL models, such as AudioMAE [23], take inspiration from MLM principles but pivot towards reconstructing the original audio content from masked segments. Given the reconstructive nature of AudioMAE pre-training, the AudioMAE features are potentially more balanced in acoustic and semantic information than those derived from only discriminative pre-training. In this work, we develop SemantiCodec with an AudioMAE encoder as one of the fundamental components.

C. Conditional Audio Generation

The development of generative models [33]–[35] has significantly propelled the field of conditional audio generation. Speech generation technologies [36]–[38] have evolved to produce speech conditioned on transcriptions and specific styles, such as speaker identity, emotion, and prosody. The generative model has also enabled novel tasks such as binaural sound synthesis [39] and synthetic speech quality refinement [40]. Meanwhile, the generation of music and sound effects has been extended to be conditioned on textual descriptions [12], [25], [41] and visual cues [42], [43], demonstrating the versatility of generative architectures.

Diffusion models [44] stand out for their exceptional generative capabilities in producing diverse and high-quality samples. Diffusion models offer a more tractable training process than GANs, emerging as a preferred choice for many researchers. To address the computational demands associated with training and inference in high-dimensional spaces, latent diffusion models (LDM) [45] have been introduced. LDMs operate on a lower-dimensional latent space derived from a VAE, significantly reducing computational complexity while maintaining the generative power of traditional diffusion models. This approach has been successfully applied in various conditional audio generation models, including AudioLDM [25], TANGO [46], AudioSR [47] and Make-An-Audio [48], illustrating the flexibility of the diffusion model. In this work, we leverage the strong generative modelling ability of LDMs to build a SemantiCodec decoder.

III. SemantiCodec

A. System Overview

As shown in Figure 2, given an input audio $\boldsymbol{x} \in \mathbb{R}^l$, where l denotes the sample length of audio, we initially transform

$\boldsymbol{x}$ into the mel-spectrogram $\boldsymbol{X} \in \mathbb{R}^{T\times F}$, with T and F indicating the temporal and frequency dimensions, respectively. Leveraging a pretrained AudioMAE $\mathcal{A}(\cdot)$, we compute the AudioMAE features $\tilde{\boldsymbol{Y}} = \mathcal{A}(X) = [\tilde{\boldsymbol{y}}_1, \tilde{\boldsymbol{y}}_2, \ldots, \tilde{\boldsymbol{y}}_L] \in \mathbb{R}^{L\times E}$, where $L = \frac{TF}{P^2}$ denotes the number of patch embedding vectors, and P and E represent the patch size and embedding size of AudioMAE, respectively. Each patch is a distinct, non-overlapping block of the mel-spectrogram processed by the AudioMAE, with multiple patches collectively forming the input to the AudioMAE encoder.

To reduce the number of patch embedding vectors, which directly influence the bitrate after quantization, we aggregate adjacent vectors of $\tilde{\boldsymbol{Y}}$ into $\boldsymbol{Y} = [\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_{\frac{L}{K}}] \in \mathbb{R}^{\frac{L}{K}\times KE}$, where $K \in \{1, 2, 4\}$ is the stack factor, yielding $\boldsymbol{y}_i = [\tilde{\boldsymbol{y}}_{iK}, \ldots, \tilde{\boldsymbol{y}}_{(i+1)K-1}]$ for $i \in \{0, 1, \ldots, \frac{L}{K} - 1\}$. Following extensive clustering on the vector $\boldsymbol{y}_i$, we derive the semantic codebook $\mathbb{E}_{\text{s}} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_{N_{\text{s}}}]$, where N_{s} denotes the number of entries in the semantic codebook. We refer to this clustering process as *semantic clustering*.

The stacked feature $\boldsymbol{Y}$ undergoes initial quantization by $\mathbb{E}_{\text{s}}$ into semantic tokens $\boldsymbol{c}_{\text{s}}$ and semantic feature $\boldsymbol{E}_{\text{s}} \in \mathbb{R}^{\frac{L}{K}\times KE}$. Subsequently, we concatenate $\boldsymbol{Y}$ and $\boldsymbol{E}_{\text{s}}$, employing an acoustic encoder $\mathcal{F}(\cdot)$ to compute the acoustic feature $\boldsymbol{Y}_{\text{A}}$, which is then quantized via an acoustic vector quantization layer with entries $\mathbb{E}_{\text{a}} \in \mathbb{R}^{N_{\text{a}}\times KE}$, outputting the acoustic tokens $\boldsymbol{c}_{\text{a}}$ and quantized acoustic feature $\boldsymbol{E}_{\text{a}}$. The final tokens for the input audio $\boldsymbol{x}$ is a merge of semantic and acoustic tokens: $\boldsymbol{c} = [\boldsymbol{c}_{\text{s}}, \boldsymbol{c}_{\text{a}}]$.

The decoder of SemantiCodec utilizes a latent diffusion model conditioned on the concatenated quantized semantic and acoustic features $\boldsymbol{E} = [\boldsymbol{E}_{\text{s}}, \boldsymbol{E}_{\text{a}}]$. The estimation of the latent diffusion model is further decoded back to waveform by a pretrained VAE decoder and a mel-spectrogram vocoder. The acoustic encoder $\mathcal{F}(\cdot)$ is joint-optimized with the acoustic codebook $\mathbb{E}_{\text{a}}$ and the latent diffusion model.

B. Semantic Clustering

AudioMAE features stand out for their ability to preserve semantic and acoustic information [49], positioning them as highly effective features for reconstruction tasks. Additionally, AudioMAE features have demonstrated strong performance in downstream classification benchmarks [23]. Given these attributes, the AudioMAE feature is selected as the input for the SemantiCodec encoder, aiming to optimize audio reconstruction quality while ensuring the retention of semantic content.

We follow AudioLDM 2 [49] for AudioMAE feature extraction. Given an audio mel spectrogram representation $\boldsymbol{X} \in \mathbb{R}^{T\times F}$, the AudioMAE first transforms $\boldsymbol{X}$ into patches of dimensions $P \times P$. These patches form the inputs to the AudioMAE encoder, which leverages a design akin to the vision transformer [50]. The output $\boldsymbol{Y}_0$ of the AudioMAE encoder has a dimension of $\frac{T}{P} \times \frac{F}{P} \times E$, which can be viewed as a sequence of tensors with length $L = \frac{TF}{P^2}$ and an embedding dimension of E. We stack the adjacent K frames of $\boldsymbol{Y}_0$ on the embedding dimension to form stacked AudioMAE feature vectors $\boldsymbol{Y} = \{\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_{\frac{L}{K}}\}$, which are used both for semantic clustering and as the input of SemantiCodec encoder.

To perform semantic quantization on the AudioMAE feature vectors $\boldsymbol{y}_i$, we utilize k-means clustering [24], a widely used technique for partitioning a dataset into clusters in which each data point belongs to the cluster with the nearest mean. Our preliminary experiment indicates that the diverse acoustic characteristics of different audio types could lead to suboptimal outcomes when a single k-means clustering model is applied to a varied audio dataset. For instance, speech may necessitate finer granularity in clustering due to its rich semantic content, unlike more homogeneous sounds such as wind or church bells. We employ an ensemble clustering approach to overcome these issues and enhance semantic clustering accuracy, motivated by the cluster ensemble approach used in HuBERT [30]. This involves training distinct k-means models for three specific audio categories: speech, music, and general sounds. The codebooks (i.e., k-means cluster centroids) derived from these domains are then combined together, forming a new ensembled codebook that accommodates the distinct acoustic features of each audio category.

C. SemantiCodec Encoder

As introduced in Section III-A, the encoding process of the SemantiCodec includes taking the stacked AudioMAE feature $\boldsymbol{Y}$ as input and calculating the tokens $\boldsymbol{c} = [\boldsymbol{c}_{\text{s}}, \boldsymbol{c}_{\text{a}}]$ and the latent features $\boldsymbol{E} = [\boldsymbol{E}_{\text{s}}, \boldsymbol{E}_{\text{a}}]$. This part introduces the derivation of $\boldsymbol{c}$ and $\boldsymbol{E}$.

1) Semantic Encoder: With the N_{s} semantic codebook centroids $\boldsymbol{e}_j \in \mathbb{E}_{\text{s}} = \{\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_{N_{\text{s}}}\}$, the semantic quantization process is given by

$$c_{\text{s}}(i) = \underset{j\in\{1,\ldots,N_{\text{s}}\}}{\arg\min} \|\boldsymbol{y}_i - \boldsymbol{e}_j\|^2, \quad \boldsymbol{E}_{\text{s}}(i) = \boldsymbol{e}_{c_{\text{s}}(i)}, \tag{1}$$

where $i \in \{0, 1, \ldots, \frac{L}{K} - 1\}$, $c_{\text{s}}(i)$ denotes the index of the closest centroid in the semantic codebook $\mathbb{E}_{\text{s}}$ to the feature vector $\boldsymbol{y}_i$, and $\boldsymbol{E}_{\text{s}}(i)$ is the centroid vector. This quantization step effectively maps each high-dimensional feature vector $\boldsymbol{y}_i$ into a discrete semantic token $c_{\text{s}}(i)$ and corresponding quantized semantic feature $\boldsymbol{E}_{\text{s}}(i)$.

2) Acoustic Encoder: While the quantized AudioMAE feature $\boldsymbol{E}_{\text{s}}$ encapsulates rich semantic information, our experiments indicate that using $\boldsymbol{E}_{\text{s}}$ alone as the condition for audio reconstruction leads to sub-optimal quality (see Table II), such as unintelligible speech. To address this, we introduce an additional acoustic encoder module with vector quantization to capture discrete acoustic detail-oriented representations. The input feature we used for acoustic quantization is calculated by an acoustic encoder $\mathcal{F}_{\Phi}(\cdot)$, which takes the stack of the AudioMAE feature before and after semantic quantization as input and outputs the acoustic feature, given by

$$\boldsymbol{Y}_A = \mathcal{F}_{\Phi}([\boldsymbol{Y}, \boldsymbol{E}_{\text{s}}]) \in \mathbb{R}^{\frac{L}{K}\times EK}, \tag{2}$$

where Φ denotes the trainable parameter in the acoustic encoder. Specifically, we employ a bi-directional long short-term memory (BiLSTM) based network [51] as the acoustic encoder. We use both $\boldsymbol{Y}$ and $\boldsymbol{E}_{\text{s}}$ as inputs because $\boldsymbol{E}_{\text{s}}$ undergoes information loss during semantic quantization, while $\boldsymbol{Y}$ retains

most of the necessary information for audio reconstruction. By providing both $\boldsymbol{Y}$ and $\boldsymbol{E}_{\text{s}}$, the second acoustic quantization layer can effectively compensate for the information lost during semantic quantization.

We use the quantization of $\boldsymbol{Y}_A$ to convert the detailed acoustic nuances into a compact, discrete format. We define the acoustic codebook as $\mathbb{E}_{\text{a}} = \{\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_{N_{\text{a}}}\}$, where N_{a} denotes the number of codebook entries and each $\boldsymbol{e}_i \in \mathbb{R}^{EK}$ represents an individual codebook vector. For each vector $\boldsymbol{y}_{a,i} \in \boldsymbol{Y}_A$, the quantization process is as follows:

$$c_{\text{a}}(i) = \underset{j \in \{1,\ldots,N_{\text{a}}\}}{\arg\min} \|\boldsymbol{y}_{a,i} - \boldsymbol{e}_j\|^2, \quad \boldsymbol{E}_{\text{a}}(i) = \boldsymbol{e}_{c_{\text{a}}(i)}, \tag{3}$$

where $i \in \{0, 1, \ldots, \frac{L}{K} - 1\}$, $c_{\text{a}}(i)$ identifies the index of the nearest centroid within the acoustic codebook $\mathbb{E}_{\text{a}}$ for the feature vector $\boldsymbol{y}_{a,i}$, and $\boldsymbol{E}_{\text{a}}(i)$ denotes the quantized vector. To encourage the acoustic encoder to produce representations that closely match the codebook entries and stabilise training, we employ a commitment loss [5], given by

$$\mathcal{L}_{\text{commit}} = \sum_i \|\boldsymbol{y}_{a,i} - \boldsymbol{E}_{\text{a}}(i)\|^2. \tag{4}$$

Following the success of HiFi-Codec [8], our approach utilizes an exponential moving average (EMA) mechanism for codebook update. The final tokens $\boldsymbol{c}$ and the representation $\boldsymbol{E}$ are the concatenation of outputs from both the semantic quantization layer and acoustic quantization layer, given by

$$\boldsymbol{c} = [\boldsymbol{c}_{\text{s}}, \boldsymbol{c}_{\text{a}}] \in \mathbb{N}^{\frac{2L}{K}}, \boldsymbol{E} = [\boldsymbol{E}_{\text{s}}, \boldsymbol{E}_{\text{a}}] \in \mathbb{R}^{\frac{L}{K} \times 2EK}. \tag{5}$$

Note that the dual-layer vector quantization architecture (semantic quantization and acoustic quantization) in SemantiCodec is different from the residual vector quantization (RVQ) approach [4], [6], where the second layer aims to refine the residual from the first VQ layer. Although both SemantiCodec and RVQ rely on multiple layers of vector quantization, SemantiCodec does not explicitly calculate residuals between each layer. Instead, SemantiCodec concatenates the outputs of each VQ layer to form the output of the encoder.

Our experiments in Section V demonstrate that the dual-layer vector quantization in SemantiCodec effectively decouple the semantic and acoustic information. Specifically, removing the acoustic quantization layer significantly degrades the reconstruction quality (Table II), while having only a minor impact on the semantic information in the latent space (Table III).

D. Latent Diffusion Model for Reconstruction

Following the success of latent diffusion models (LDMs) for conditional audio generation [25], [47], [48], we employ an LDM as the decoder to reconstruct the original audio x. The LDM models the data distribution in a latent space constructed from a VAE, which is directly adopted from the VAE used in AudioLDM 2 [49]. Note that the VAE encoder is only used during SemantiCodec training, and is discarded during SemantiCodec inference. Compared with the original diffusion model [52], the high-dimensional spectrogram $\boldsymbol{X}$ is compressed to the low-dimensional latent $\mathbf{z}_0$ in the LDM to significantly alleviate the computations. Then a diffusion model is trained to gradually generate $\mathbf{z}_0$ from Gaussian noise. The forward diffusion process comprises a series of N Markov transition steps that gradually transform $\mathbf{z}_0$ into a Gaussian distribution by noise injection. The forward step $n-1$ is defined as:

$$q(\mathbf{z}_n|\mathbf{z}_{n-1}) = \sqrt{1-\beta_n}\mathbf{z}_{n-1} + \sqrt{\beta_n}\epsilon_n, \tag{6}$$

where β_n is the predefined noise schedule. By compositing these forward steps, we can derive the closed-form distribution of an arbitrary step n given the initial $\mathbf{z}_0$ [52]:

$$q(\mathbf{z}_n|\mathbf{z}_0) = \sqrt{\bar{\alpha}_n}\mathbf{z}_0 + \sqrt{1-\bar{\alpha}_n}\epsilon_n, \tag{7}$$

where $\alpha_n = 1 - \beta_n, \bar{\alpha}_n = \prod_{n=1}^{n} \alpha_n$ and $\epsilon \sim \mathcal{N}(0, I)$. With enough diffusion steps N, $q(\mathbf{z}_n)$ will approximate a standard Gaussian distribution $\mathcal{N}(0, I)$. The LDM is trained to model the reverse probability $p_\theta(\mathbf{z}_{n-1}|\mathbf{z}_n, \boldsymbol{E})$ conditioned on the SemantiCodec encoder output $\boldsymbol{E}$. Although $\boldsymbol{E}_{\text{a}}$ may already contain part of the information from $\boldsymbol{E}_{\text{s}}$, we feed both into the diffusion model as conditions to enable both quantization layers to collaboratively capture as much information from $\boldsymbol{Y}$ as possible.

Recent research by Lin et al. [53] has identified limitations in the prevalent noise scheduling techniques employed in diffusion models, specifically highlighting that the noisy latent at the last forward diffusion step $\mathbf{z}_N$ fails to follow a Gaussian distribution. To rectify this discrepancy, we adopt the strategy proposed by Lin et al. [53] and implement a cosine noise schedule. This modification guarantees the attainment of a standard Gaussian distribution in the last step of the diffusion process during training, thereby enhancing the consistency of the LDM between training and inference. To improve the generation performance and stabilize the sampling process, we switch the standard noise prediction objective to velocity prediction proposed in [54]. The LDM training loss [54] can be formulated as:

$$\boldsymbol{v}_n = \sqrt{\bar{\alpha}_n}\epsilon - \sqrt{1-\bar{\alpha}_n}\mathbf{z}_0, \tag{8}$$

$$\mathcal{L}_{\text{recon}} = \|\boldsymbol{v}_n - \mathcal{G}_\theta(\mathbf{z}_n, n, \boldsymbol{E})\|^2, \tag{9}$$

where $\mathcal{G}_\theta$ denotes the LDM and θ is the set of trainable parameters.

For model inference, we adopt the Denoising Diffusion Implicit Models (DDIM) sampler [55], which uses non-Markovian diffusion processes for sampling, allowing for computationally efficient sampling while maintaining high generation quality. Finally, the audio $\hat{\boldsymbol{x}}$ is reconstructed via the pretrained VAE decoder and a vocoder. The vocoder is a HiFi-GAN-based architecture [56] and is directly adopted from the pretrained AudioLDM 2. Our LDM adopts the Transformer-UNet (T-UNet) introduced in AudioLDM 2 [49], which incorporates self-attention and cross-attention Transformer blocks between convolutional blocks, significantly enhancing the model capacity for complex audio patterns. However, our experiments suggest that the impact of T-UNet parameter numbers, such as 75 million or 346 million, on the quality of audio reconstruction is relatively minor compared to their

role in text-to-audio generation tasks. Therefore, we adopt a T-UNet with reduced parameter size compared with AudioLDM 2.

To achieve potentially better quality in reconstruction, we adopt classifier-free guidance (CFG) [57], [58], a common approach to guiding the audio generation in diffusion models. The condition $\boldsymbol{E}$ in Equation (9) is randomly discarded with a certain probability during training so that both conditional generation models $v_\theta(\mathbf{z}_n, n, \boldsymbol{E})$ and unconditional generation models $v_\theta(\mathbf{z}_n, n)$ are optimized in a multi-task paradigm. During sampling, the original $v_\theta(\mathbf{z}_n, n, \boldsymbol{E})$ is complemented by the weighted combination of velocities predicted by conditional and unconditional models:

$$(1 - w) \cdot v_\theta(\mathbf{z}_n, n, \boldsymbol{E}) + w \cdot v_\theta(\mathbf{z}_n, n), \tag{10}$$

where w is the guidance scale. We show the effect of different CFG guidance scales in Figure 6.

E. Training Objective

As shown in Figure 2, we keep the pretrained AudioMAE, VAE and vocoder parameters frozen. The k-means semantic clustering centroids are separately obtained before the training of the latent diffusion model and are also kept frozen. The acoustic encoder, acoustic vector quantization layer and the latent diffusion model are jointly optimized by a sum of the reconstruction loss and the commitment loss in the acoustic vector quantization layer, denoted by

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{commit}}. \tag{11}$$

IV. Experimental Setup

A. Datasets

1) Training Datasets: Our model training is supported by various datasets, which can be generally classified into three categories: speech, music, and general sounds. For speech, we utilize the GigaSpeech (GGS) [59] dataset, a comprehensive English speech recognition corpus with around $10,000$ hours of transcribed audio, and the speech dataset collected by VoiceFixer [60] on OpenSLR[1], featuring a multi-lingual speech dataset with $186,514$ short audio clips. The music category includes the Million Song Dataset (MSD) [61], which provides a vast collection of $510,000$ music tracks with metadata. We also adopt datasets including MedleyDB [62], and MUSDB18 [63] training subset, mostly used for music source separation [64]. For general sounds, we engaged with AudioSet (AS) [65], the largest classification dataset offering two million ten-second audio clips across 527 categories, WavCaps [66] that includes ChatGPT-assisted weakly-labeled audio captions for $403,050$ clips, and VGGSound (VS) [67], a substantial audio-visual dataset with around $190,000$ videos from which we only utilized the audio data. All the audio data are in 16 bits and resampled to 16 kHz during training and evaluation. Therefore, all SemantiCodec configurations presented in this work operate at a 16 kHz sampling rate. We plan to explore higher sampling rate versions of SemantiCodec in future research. Our ablation studies in Section V-C2 and Section V-C3 utilize only 10% of the training data to speed up the model training.

2) Reconstruction Performance Evaluation: To assess the reconstruction capabilities of our audio codec, we carefully select and evaluate three distinct categories of datasets similar to building the training set. Within the speech category, we choose a subset of the LibriTTS clean test set [68], selecting 300 speech utterances randomly. These utterances, each lasting between 8 and 10 seconds, come with detailed transcription annotations, offering a rich basis for evaluating speech reconstruction accuracy. We randomly select 500 segments from the AudioSet evaluation set for general sound evaluation, ensuring a diverse representation of ambient sounds, effects, and non-musical content. Our music data evaluation leverages the MUSDB18 test set [63], comprising 50 songs with isolated tracks for vocals, drums, bass, and other elements. From each of the four tracks and their mixture for every song, we randomly select a 10-second segment, ensuring it is non-silent, to form our evaluation set. Our evaluation dataset encompasses 1050 audio samples, achieving a relatively balanced distribution across speech, music, and general sounds. Our MUSHRA (MUltiple Stimuli with Hidden Reference and Anchor) test [69] is performed on the same evaluation dataset while only 10% of the data is used to save effort for subjective evaluation. Our evaluation set and metrics implementations are publicly available[2].

3) Semantic Information Evaluation: To assess the richness of semantic information captured by audio codec representations, we employ a diverse array of datasets following a subset of the Holistic Evaluation of Audio Representations (HEAR) benchmark [22]. We choose the tasks oriented towards audio classification [70], eliminating tasks like Gunshot Triangulation, which aims to recognize recording devices. Also, for convenience, we choose the datasets where the duration of each audio clip is shorter or close to 10 seconds. The evaluation datasets we employed include the following: (i) NSynth Pitch (NSPitch) [71] is utilized to evaluate the ability to recognize musical pitches, a fundamental aspect of music theory and auditory perception. (ii) ESC-50 [72] encompasses a broad array of environmental sounds such as rainstorms and animal calls, testing the versatility of the codec features towards general sound effects. (iii) LibriCount (LbCount) [73] is the dataset used to determine the number of speakers in an audio clip, combining speech detection with the differentiation of speakers within complex auditory scenes. (iv) CREMA-D (CRM-D) [74] focuses on speech emotion recognition, requiring models to classify a wide spectrum of emotional states conveyed through speech. (v) Vocal Imitations (VoImit) [75] dataset examines the ability to classify non-verbal human vocal imitations of various sounds. (vi) Speech Commands (SC) [76] dataset tests the recognition of specific spoken commands, emphasizing speech clarity and command accuracy. These audio classification datasets collectively offer a comprehensive evaluation framework, spanning musical notes, environmental sounds, speech nuances, and non-verbal vocalizations, to

[1]https://openslr.org/

[2]https://zenodo.org/records/11047204

thoroughly assess the semantic capabilities of audio codec representations.

B. Baselines

We employ several state-of-the-art neural audio codecs as baselines for comparison, including Encodec (EC), Descript Codec (DAC), and HiFi-Codec (HC), which have demonstrated success in the domain of general audio. The open-sourced Encodec[3] is trained on a variety of audio datasets sampled at 24 kHz and can compress audio to 1.5, 3.0, 6.0, 12.0, and 24.0 kbps. Since SemantiCodec operates at a relatively low bitrate, we only include Encodec at 1.5, 3.0 and 6.0 kbps for comparisons. For DAC, we adopt the open-sourced 6.0 kbps model that operates on 16 kHz as one of the baselines. As DAC does not have checkpoints open-sourced for lower bitrates, we train three new versions of DAC on the same training data as SemantiCodec following the setting described in their paper [7]. The three DAC settings have a bit rate of 0.47, 0.78, and 1.41 kbps, which are comparable to the three variants of SemantiCodec. For HiFi-Codec, we adopt the 2.0 kbps checkpoint[4]. The baseline models we used, including Encodec, Descript Codec, and HiFi-Codec, contain approximately 23, 74, and 63 million parameters, respectively.

C. Evaluation Metrics

1) Reconstruction Performance Evaluation: To assess the reconstruction quality of audio codecs with objective metrics, we employ mel spectrogram distance (MEL), short-time Fourier transform distance (STFT), and the virtual speech quality objective listener score (ViSQOL) [77]. MEL and STFT metrics quantify spectrogram discrepancies, with STFT providing a more nuanced capture of high-frequency fidelity. Our implementation for MEL and STFT follows the approach detailed in DAC [7], employing multiple window lengths to accommodate various audio signal types effectively. ViSQOL, a full-reference and intrusive perceptual quality metric, is designed to estimate the subjective mean opinion score (MOS) of audio quality. ViSQOL offers two modes for evaluation: an audio mode for 48 kHz samples and a speech mode for 16 kHz samples. Considering our focus on 16 kHz audio codecs, we resample both the ground truth and reconstructed audio to 48 kHz to leverage the audio mode for evaluations on the AudioSet and MUSDB18 datasets. Additionally, to evaluate the intelligibility of speech signals, we employ the word error rate (WER), a critical metric for evaluating the performance of automatic speech recognition systems by calculating the percentage of errors in the form of substitutions, deletions, or insertions relative to a reference transcript. We utilize the *whisper-large-v3* [78][5] model to transcribe our reconstruction and remove all punctuations in the original LibriTTS transcriptions before calculating WER.

[3]https://github.com/facebookresearch/encodec
[4]https://github.com/yangdongchao/AcademiCodec
[5]https://huggingface.co/openai/whisper-large-v3

2) Semantic Information Evaluation: We assess the semantic distinctiveness of the audio codec representation by analyzing classification accuracy on the datasets described in Section IV-A, following the methodology outlined in the HEAR evaluation benchmark [22]. Our evaluation specifically focuses on the quantized output of the encoder, which serves as input to the decoder. The codec model is frozen to extract the quantized features without fine-tuning. The feature sequences are averaged along the time axis to obtain the clip-level features, which represents the global information of an audio clip. A shallow downstream multi-layer perceptron (MLP) classifier with two linear layers is trained on clip-level features. The classification performance is reported to indicate the semantic distinctiveness of the codec. Furthermore, several studies on audio language models use powerful auto-regressive (AR) language models to predict tokens from the first VQ layer while leaving the rest of the tokens to be predicted by non-AR language models [15]. Therefore, we also report the classification performance using tokens from the first VQ layer to assess the semantic richness.

3) MUSHRA Test: To assess the subjective audio reconstruction quality of various audio codecs, we adhere to the established MUSHRA test protocol [69]. Participants are required to compare open audio samples against corresponding hidden references, predefined anchors, and the output of different systems. Ratings are assigned on a scale from 0 to 100. For our MUSHRA evaluation, we randomly select 10% of the audio from our evaluation set, comprising 25 music tracks, 30 speech recordings, and 50 general sound samples. Participants are tasked with rating the audio quality of nine different samples, which include three variations each of SemanticCodec and Descript codecs at different bitrates, an open-source Encodec at 1.5 kbps, the original audio (ground truth), and an anchor of the original audio with a low-pass filter applied at 3.5 kHz. Note that the original MUSHRA test requires two anchor samples with cut-off frequency of 3.5 kHz and 7 kHz. However, since our test samples contain frequency information only up to 8 kHz, the 7 kHz anchor would not offer sufficient differentiation and was therefore omitted as an anchor in our test. We ensure each set of audio receives evaluations from at least 10 raters. Participants are instructed with *This task evaluates the quality proximity between an audio sample and its reference. Please listen carefully to the reference audio and then rate the quality of each test audio clip compared to the reference. Use the scale where* 0 *indicates no resemblance to the reference, and* 100 *means perfectly the same as the reference.* Including deliberately degraded anchors allows the MUSHRA test to reveal subtle perceptual distinctions and quality variances across codecs. We perform post-screening following the standard MUSHRA test protocol [69]. Our MUSHRA test has received a favourable opinion from the ethics review through completing the University of Surrey self-assessment governance and ethics form. We calculate and report the mean MUSHRA score for each codec as its final subjective evaluation metric.

D. Implementation Details

We implemented the k-means clustering on the high-dimensional AudioMAE features extracted from our training dataset mentioned in Section IV-A. This procedure was carried out individually for three types of audio: music, speech, and general sound. For each type of audio, we run the clustering with different numbers of centroids, including 1024, 2048, 4096, 8192, and 16384, to produce 15 distinct sets of centroids. To manage the computational complexity of clustering high-dimensional features, we made several optimizations to the algorithm, which is available online[6].

In building the final semantic codebook for SemantiCodec, we combined codebooks from each of the three audio domains. Given that general sound tends to encompass a broader array of sounds than speech and music, we assigned twice as many centroids. This created four distinct semantic codebooks with varying number of centroids: 4096, 8192, 16384, and 32768. For example, the codebook with 16384 centroids includes 8192 centroids for general sound and 4096 centroids each for speech and music. This selection and combination process ensures our SemantiCodec has a comprehensive semantic codebook that accurately captures the diverse range of audio it processes. We repeat the above clustering and merging process for three different settings of stack factors $K \in \{1, 2, 4\}$. We randomly select and employ one of the four semantic codebooks in each training batch during model optimization. Therefore, our model supports variable vocabulary sizes and bitrates. Table I summarizes the available settings of SemantiCodec with the corresponding codebook sizes, bit rates, and token rates. The acoustic codebook utilizes a fixed-size codebook. We employ 32768 centroid semantic codebook and 8192 centroid acoustic codebook by default during model evaluation. Our experiment result indicates that more semantic centroids can moderately improve the reconstruction quality (see Figure 7).

TABLE I
BIT RATE ALLOCATION TABLE FOR SEMANTIC QUANTIZATION LAYER (SEM) AND ACOUSTIC QUANTIZATION LAYER (ACO).

Token/Sec (SEM+ACO)	Codebook Size SEM	Codebook Size ACO	Kbps SEM	Kbps ACO	Overall kbps
100	32768	8192	0.75	0.65	1.40
	16384		0.70		1.35
	8192		0.65		1.30
	4096		0.60		1.25
50	32768	8192	0.375	0.325	0.700
	16384		0.350		0.675
	8192		0.325		0.650
	4096		0.300		0.625
25	32768	8192	0.1875	0.1625	0.3500
	16384		0.1750		0.3375
	8192		0.1625		0.3250
	4096		0.1500		0.3125

We utilize the AudioMAE model pretrained on AudioSet[7], which features an embedding dimension of 768. The resulting feature dimension is $K \times 768$ when stacking adjacent frames. Our implementation strategy for both the AudioMAE encoder and the diffusion decoder aligns with the approach described in AudioLDM 2. A notable modification in our setup is the adoption of a more compact T-UNet architecture with a base channel number of 64 and a parameter number of 75 million, in contrast to the larger version in AudioLDM 2 which features a base channel number of 128 with 346-million parameters. SemantiCodec is configured to operate at three different token rate settings, including 25, 50, and 100 tokens per second. For each setting, the model undergoes training for $500,000$ steps on the designated training set with two A100-Amphere-80GB GPUs with a batch size of 48. With a basic learning rate of 10^{-4}, we incorporate a linear warm-up phase over the first $5,000$ steps. In the T-UNet architecture, the output from the SemantiCodec encoder, denoted as $\boldsymbol{E}$, is integrated via cross-attention mechanisms. Given that $\boldsymbol{E}$ lacks inherent positional information, we enrich it with fixed positional embeddings before input into the cross-attention layers.

SemantiCodec is trained on audio segments exactly 10.24 seconds in length. To accommodate audio files of varying lengths, we have developed solutions involving either trimming or an overlap-add approach. For audio files shorter than 10.24 seconds, we pad them to the required duration, compute the token encoding, and then remove any tokens corresponding to the padded region. For files longer than 10.24 seconds, we segment the audio into 10.24-second chunks without overlap, compute their token embeddings, and concatenate these segments. During inference, the decoder uses a window length of 10.24 seconds with a 6.25% overlap between consecutive windows. The audio content from overlapping segments is blended using a linear decay and gain before combining.

V. RESULTS

A. Reconstruction Quality

Table II shows the audio reconstruction quality of SemantiCodec and baselines described in Section IV-B indicated by objective metrics. We use *Token/Sec* to denote the number of tokens the audio signal is encoded into per second, which is crucial in audio language modelling as it influences the length of the audio sequence. Given the variability in semantic codebook sizes, models with identical token rate may yield different bit rates (see Figure 7), denoted by *kbps*. SemantiCodec significantly outperforms DAC on the reconstruction quality at similar bitrates. With as low bitrate as 0.70 kbps, SemantiCodec still outperforms the 1.5 kbps Encodec on reconstruction quality. It is remarkably comparable to HiFi-Codec operating at 2.0 kbps, as the average ViSQOL score indicates. With an ultra-low bitrate of 0.36 kbps, SemantiCodec still achieves a better ViSQOL score than the 1.41 kbps DAC. The 1.40 kbps SemantiCodec demonstrates performance on par with the 3.0 kbps Encodec. The superior performance of SemantiCodec in reconstruction quality at low bitrates indicates its potential in efficient audio transmission, storage and audio-based language modelling since it provides shorter discrete representations of audio without substantially compromising the reconstruction quality.

[6]https://github.com/haoheliu/kmeans_pytorch
[7]https://github.com/facebookresearch/AudioMAE

TABLE II
OBJECTIVE EVALUATION OF SEMANTICODECS AND COMPETING BASELINE CODECS AT VARIOUS BITRATES ON SPEECH, MUSIC AND GENERAL SOUND. VIS STANDS FOR THE VISQOL METRIC.

			General Sound			Music			Speech				**Average**
Model	Kbps	Token/Sec	MEL↓	STFT↓	VIS↑	MEL↓	STFT↓	VIS↑	MEL↓	STFT↓	VIS↑	WER↓	VIS↑
GroudTruth	–	-	0.0	0.0	4.99	0.0	0.0	4.99	0.0	0.0	4.99	2.09	4.99
SemantiCodec w. GT AudioMAE	–	-	3.78	3.89	4.58	3.79	3.40	4.56	3.77	3.18	4.71	2.7	4.61
SemantiCodec w.o. Acoustic VQ	0.70	50	7.29	4.93	2.43	7.67	4.44	2.61	8.68	4.58	2.78	55.6	2.61
DAC	6.00	600	2.91	3.03	4.36	2.83	2.90	4.54	2.79	2.92	4.71	3.0	4.54
Encodec	6.00	600	4.38	4.10	4.00	4.17	2.90	4.14	4.54	3.19	4.35	3.3	4.16
	3.00	300	4.84	4.26	3.58	4.67	3.11	3.78	5.06	3.40	4.10	3.7	3.82
	1.50	150	5.39	4.47	3.04	5.30	3.33	3.27	5.83	3.67	3.67	5.0	3.33
HiFi-Codec	2.00	200	4.35	3.61	3.11	4.37	3.11	3.42	3.93	2.99	4.18	3.6	3.57
DAC	0.47	47	7.56	4.58	2.12	7.80	4.24	2.33	8.62	4.70	2.73	28.2	2.39
	0.78	78	6.73	4.41	2.47	6.44	3.88	2.80	6.76	4.13	3.19	11.6	2.82
	1.41	141	6.56	4.78	2.85	6.30	3.72	3.10	6.71	3.91	3.44	5.0	3.13
SemantiCodec	0.36	25	5.06	4.02	2.84	5.22	3.76	3.18	5.77	3.72	3.49	19.6	3.17
	0.70	50	4.67	3.97	3.20	4.74	3.62	3.54	4.95	3.49	3.92	5.1	3.55
	1.40	100	4.39	3.79	3.48	4.44	3.56	3.80	4.54	3.38	4.17	3.4	3.81

Using the ground truth unquantized AudioMAE features as the condition for the latent diffusion model, the setting SemantiCodec *w. GT AudioMAE* marks the performance upper bound of SemantiCodec. SemantiCodec *w. GT AudioMAE* demonstrates strong performance on the reconstruction quality with an average ViSQOL score of 4.61, which indicates that the original AudioMAE features contains sufficient information to reconstruct the audio signal. While DAC and Encodec directly use MEL and STFT as training objectives, potentially enhancing their performance on these metrics, SemantiCodec, despite not doing the same, achieves superior results at comparable bitrates, highlighting its overall better performance.

By removing the second VQ layer, as shown in the setting of SemantiCodec *w.o. Acoustic VQ*, the model performance exhibit a considerable degradation, with a WER of 55.6 and a ViSQOL of 2.61, highlighting the importance of the second acoustic VQ layer for acoustic reconstruction. Finally, comparing the reconstruction performance across different audio types, it can be concluded that general sound consistently exhibits a slightly inferior reconstruction quality than music and speech. The challenges in reconstructing general sound signals may be attributed to the intrinsic complexity and variability of general sound content. This also validates our choice of assigning more semantic centroid numbers for general sound than speech and music in Section IV-D.

Figure 3 showcases the reconstruction performance in terms of subjective MUSHRA scores. We only incorporate codecs with a bitrate of less than or equal to 1.5 kbps into evaluation since we aim at developing codecs with ultra-low bitrates. The result is consistent with the objective evaluation shown in Table II that SemantiCodec significantly outperforms counterparts with a similar or even higher bitrate. With the reference audio being scored to 93.9, the 1.5 kbps Encodec model achieves a 47.8 average MUSHRA score, while even our 0.35 kbps SemantiCodec achieves a significantly higher score of 55.7. The high MUSHRA score of SemantiCodec can be attributed to the generative nature of the model, which leads

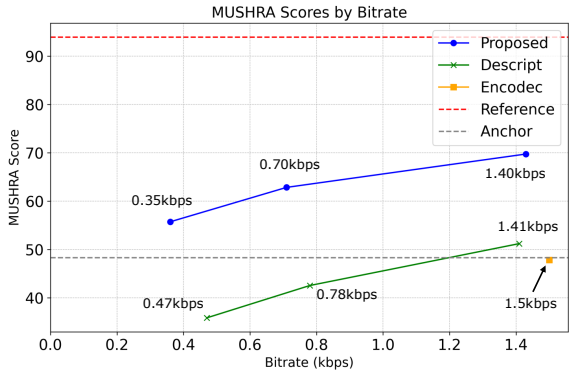

Fig. 3. The average MUSHRA test score on our curated evaluation set. Our proposed SemantiCodec outperforms the baseline models even with a significantly lower bitrate. For fair comparisons, DAC and HiFi-Codec are not included as they use much higher bit rates.

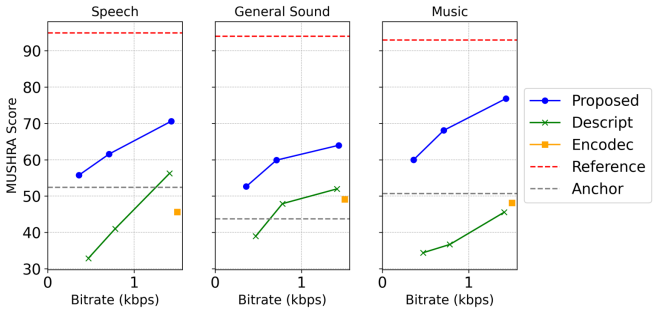

Fig. 4. The MUSHRA test score on different domains.

to better audio quality at ultra-low bitrate. By comparison, we observe strong artifacts at similar bitrates on DAC and Encodec (shown in Figure 5), which can significantly impact the perceived quality for the listener. Figure 4 provides further analysis of the MUSHRA score on different domains of audio, which shows similar trends as Figure 3. SemantiCodec demonstrates significantly better performance in music, typically involving more complex acoustic information. This suggests

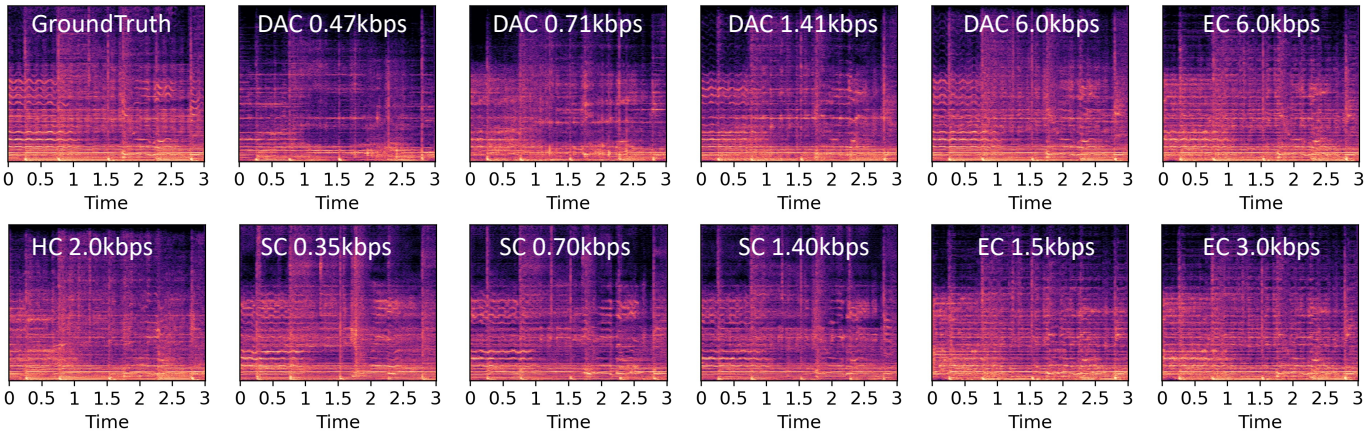

Fig. 5. The log-STFT spectrogram of the ground truth audio and the reconstruction audio with different audio codecs. DAC, EC, HC, and SC are the descript codec, Encodec, HiFi-Codec, and SemantiCodec, respectively.

TABLE III
SEMANTIC EVALUATION OF SEMANTICODECS AND COMPETING BASELINE CODECS USING THE HEAR BENCHMARK.

VQ setting	Model	Kbps	Token/Sec	ESC-50 (%)	NSPitch (%)	SPC (%)	LbCount (%)	CRM-D (%)	VoImit (%)	Average (%)
Unquantized	AudioMAE	-	-	79.5	82.0	48.0	69.4	67.3	17.4	60.6
All VQ layers	Encodec	6.00	600	40.7	60.8	27.3	45.0	44.2	4.4	37.1
	HiFi-Codec	2.00	200	36.3	71.0	26.5	58.2	45.6	4.3	40.3
	DAC	6.00	600	33.4	56.1	21.0	46.4	39.9	3.4	33.4
		1.41	141	41.1	**80.9**	30.3	59.5	44.7	4.8	43.5
		0.78	78	39.5	78.5	30.3	59.5	45.7	5.0	43.0
		0.47	47	36.7	75.7	26.4	59.6	44.5	4.9	41.3
	SemantiCodec	1.40	100	**63.8**	73.3	**43.6**	67.0	**57.9**	**9.6**	**52.5**
		0.70	50	60.9	64.9	41.9	**71.6**	53.2	9.3	50.3
		0.35	25	56.4	61.3	33.7	70.4	46.9	8.0	46.1
First VQ Layer	Encodec	0.75	75	32.0	45.3	23.0	44.8	40.7	4.2	31.6
	HiFi-Codec	1.00	100	33.7	58.9	25.9	58.3	44.3	4.1	37.5
	DAC (6.00k)	0.5	50	23.8	23.3	15.9	43.1	38.4	3.1	24.6
	DAC (1.41k)	0.16	16	29.0	44.2	19.9	57.0	39.5	4.0	32.3
	DAC (0.78k)	0.16	16	27.6	39.9	17.9	56.6	40.9	4.0	31.1
	DAC (0.36k)	0.16	16	29.7	46.9	18.3	58.2	43.0	4.0	33.3
	SemantiCodec	0.70	50	**66.6**	**73.9**	**42.7**	**66.7**	**57.5**	**11.1**	**53.1**
		0.36	25	64.4	70.2	36.0	65.0	54.1	10.7	50.1
		0.18	13	59.6	66.3	30.7	61.3	45.8	9.8	45.6

that SemantiCodec is highly effective at handling intricate scenarios.

B. Semantic in the Codec Tokens

The semantic richness of different codecs is demonstrated in Table III. First, when all VQ layers are utilized, SemantiCodec significantly outperforms baseline models in semantic information. Notably, even at a low bitrate of 0.35 kbps, the semantic performance of SemantiCodec surpasses that of higher bitrate counterparts like the 6.0 kbps Encodec and DAC. This indicates that AudioMAE features serve as effective means of semantic encoding.

Then, comparing codecs at different bitrates below 1.5 kbps, both DAC and SemantiCodec exhibit a decline in semantic performance as the bitrate decreases. However, even at the lowest bitrate of 0.35 kbps, SemantiCodec outperforms DAC at 1.41 kbps. Interestingly, despite the inferior reconstruction quality of DAC at lower bitrates, the semantic richness retained at low bitrates is significantly better than that of 6.0 kbps. This may be attributed to the information bottleneck imposed on the encoded representations during training with low bitrates. The model must preserve coarse audio contents in representations while discarding acoustic details, making the classifier easier to train on downstream tasks.

Since many studies utilize tokens from the first VQ layer to train audio language models, we investigate the semantic information in tokens under this situation. As shown in the lower part of Table III, the baseline codecs exhibit a substantial drop in semantic performance when switching from evaluating all VQ layers to the first layer only. For example, the first VQ layer of the 6.0 kbps DAC only achieves an average accuracy of 24.6. In contrast, SemantiCodec maintains similar semantic accuracy with only the first VQ layer. This supports our assumption that semantic information is primarily encoded by semantic codes obtained through k-means clustering of AudioMAE features, with acoustic details augmented by subsequent acoustic codes. However, there is still a performance gap between unquantized AudioMAE features (ACC 60.6) and the quantized latent space of SemantiCodec (ACC 53.1), highlighting the need for further research into mitigating the loss caused by quantization.

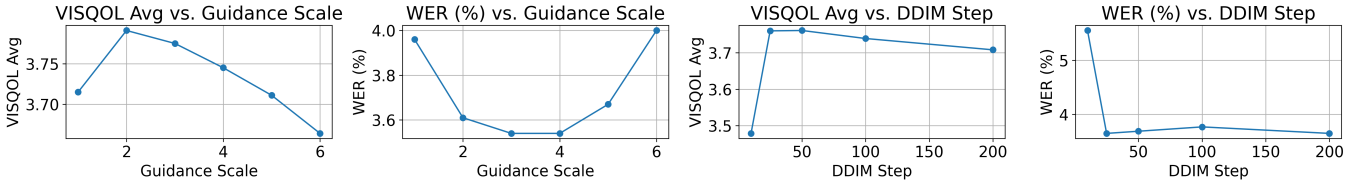

Fig. 6. The impact of CFG guidance scale and DDIM sampling steps on reconstruction quality.

TABLE IV
UTILIZING A VARIABLE SEMANTIC CODEBOOK SIZE IS BENEFICIAL FOR THE RECONSTRUCTION QUALITY, COMPARED WITH USING A FIXED SEMANTIC CODEBOOK OF SIZE 32768.

	fixed vocab size		variable vocab size	
Stack Factor	ViSQOL-Avg	WER	ViSQOL-Avg	WER
$K = 1$	3.78	3.60	3.81	3.40
$K = 2$	3.41	6.30	3.55	5.10
$K = 4$	3.04	21.8	3.17	19.6

TABLE V
THE IMPACT OF K-MEANS CENTROID NUMBER ON THE SEMANTIC RICHNESS OF SEMANTICODEC.

K-means Centroids	ACC
4096	49.0
8192	50.9
16384	52.1
32768	**53.1**

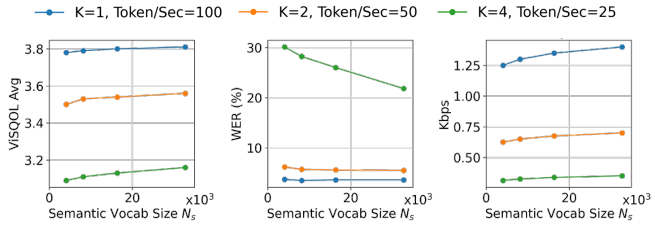

Fig. 7. The impact of semantic vocabulary size on average ViSQOL, WER, and bitrate.

TABLE VI
THE IMPACT OF SEMANTICODEC ENCODER PARAMETERS ON THE RECONSTRUCTION QUALITY.

Acoustic Codebook Size	LSTM Bi-directional	VISQOL Avg	WER (%)
8192	✓	**3.602**	**4.57**
4096	✓	3.583	5.05
2048	✓	3.550	4.66
8192	✗	3.596	4.80

C. Ablation Study

1) Variable Semantic Codebook Size: As detailed in Section IV-D, during training, we employ a variety of semantic codebooks, enabling SemantiCodec to accommodate different vocabulary sizes. To explore the efficacy of this approach, we conduct experiments comparing a fixed vocabulary size against a variable one, maintaining an acoustic codebook size of 8192 in both scenarios. We also investigate different stack factors K to assess their impact on performance. As presented in Table IV, utilizing a variable vocabulary size significantly enhances the average ViSQOL score and reduces the WER. This improvement likely stems from the exposure to diverse codebooks during training, which enhances the ability of the model to interpret and process the quantized AudioMAE features effectively.

We also study the effect of semantic codebook size on semantic richness by calculating classification accuracy with the quantized AudioMAE features. Table V shows as the centroid number increases, the quantization error induced by k-means modelling is reduced, hence the semantic performance consistently improves. Similar behaviour can be observed in the reconstruction performance, as shown in Figure 7. When using a larger number of k-means centroids (i.e., semantic vocabulary size), reconstructed speech quality steadily improves. However, as also shown by Figure 7, higher semantic vocab size will impact the bitrate, posing a trade-off between reconstruction quality and bitrate.

2) Acoustic Representation Learning: The size of the acoustic codebook determines the granularity of the trainable vector quantization module. Analogous to the influence of k-means centroid numbers, larger codebook size leads to better reconstruction quality, as indicated by Table VI.

The complexity of the LSTM acoustic encoder influences its capacity to refine and augment quantized AudioMAE features. Replacing the original bi-directional LSTM with a unidirectional one results in a significant drop in speech reconstruction quality, as indicated by WER. This validates the necessity of the bidirectional connection in the LSTM for efficiently extracting and encoding the contextual information and dependencies of AudioMAE feature sequences.

3) Learnable Semantic Codebook: The semantic codebook in SemantiCodec is frozen during model training. To validate the importance of performing k-means clustering beforehand, we conduct another experiment by replacing the semantic

TABLE VII
THE IMPACT OF REPLACING THE K-MEANS CENTROID WITH A LEARNABLE VECTOR QUANTIZATION LAYER.

SemantiCodec	ViSQOL ↑	WER ↓	ACC ↑
Original	**3.60**	**4.57**	**53.1**
Learnable Semantic VQ	3.57	5.10	28.5

codebook with a learnable VQ layer, similar to the acoustic VQ layer. As shown in Table VII, even though the reconstruction performance of SemantiCodec is not largely affected without the use of k-means centroids, the semantic evaluation accuracy shows a notable decrease from 53.1% to 28.5%. This indicates the validity of k-means centroids used in the semantic codebook for maintaining rich semantic information in audio tokens. Directly training a codec model with two VQ layers results in the codec neglecting semantic information and focusing solely on reconstruction capabilities, as the training loss function is limited to reconstruction loss.

4) DDIM Sampling Setups: As shown in Figure 6, the guidance scale w (see Equation (10)) in CFG can influence the reconstruction quality. CFG does not provide enough condition-oriented guidance through unconditional generation probability when the scale is too small. Conversely, when the scale is too large, the effect of the condition is diluted by the emphasis towards unconditional generation probability. Therefore, w with values that are too small or too large leads to unsatisfactory reconstruction quality. The best reconstruction quality is achieved under a moderate scale near 3.0.

The sampling step also plays a crucial role in the reconstruction quality of our latent diffusion decoder. The reconstruction quality of SemantiCodec is moderate, with a small number of sampling steps, e.g., ten steps. With a number of sampling steps larger than 25, the reconstruction quality significantly improves. This is promising for rapid sampling during audio reconstruction with tokens from SemantiCodec.

VI. Conclusion

In this study, we have proposed SemantiCodec, an audio codec which can be applied to diverse audio types with ultra-low bitrates and rich semantic information in the audio tokens. SemantiCodec supports several bitrates between 0.31 kbps and 1.40 kbps. With a semantic and acoustic decoupled architecture, SemantiCodec achieves effective compression without significantly sacrificing quality with token rates of 50 and 100 per second. At an ultra-low rate of 25 tokens per second, SemantiCodec still demonstrates significantly strong audio quality compared with state-of-the-art audio codecs. Our experiment result also shows that the semantic information within the SemantiCodec tokens is significantly richer than in previous neural audio codecs.

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